



Motivation

Real-world perception faces continual shifts in domains and sensing modalities.

Models must adapt to new conditions without forgetting previously learned ones.



Changing Domains and Modalities

Continual Learning Trade-offs

Approach	Stability	Plasticity	Scalability
Fine-Tuning	✗	✓	✓
Regularization	◐	◐	✓
Full Network Experts	✓	✓	✗
MILE	✓	✓	✓

MILE combines expert specialization with parameter-efficient scalability.

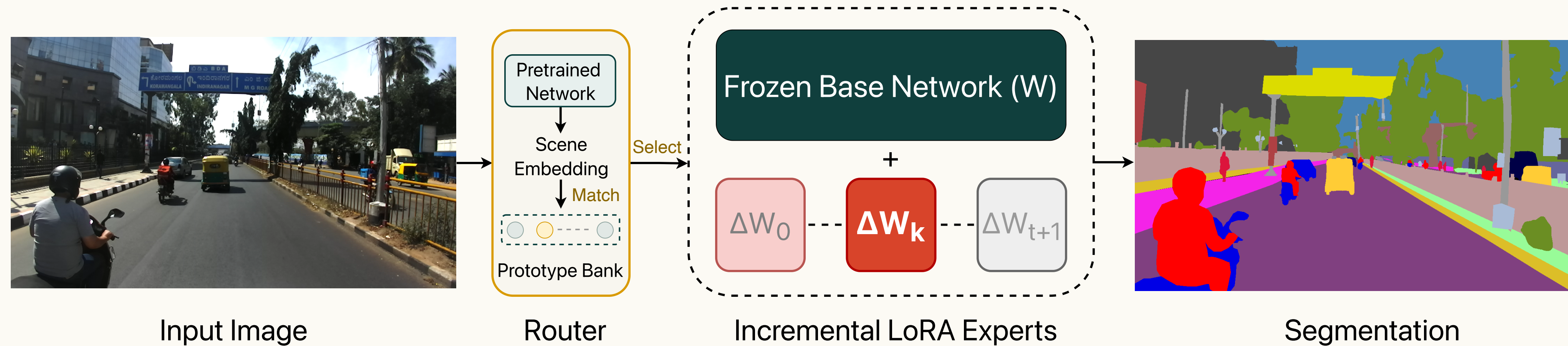
MILE Overview

UNIFY: One framework for continual semantic segmentation across domain and modality shifts.

SPECIALIZE: LoRA experts capture shift-specific knowledge while the backbone remains frozen.

ROUTE: Prototype-based routing selects the appropriate expert at inference - No task ID required.

MILE: Mixture of Incremental LoRA Experts



TRAINING

Freeze the backbone. Compute a prototype and learn one low-rank ΔW expert per new task.

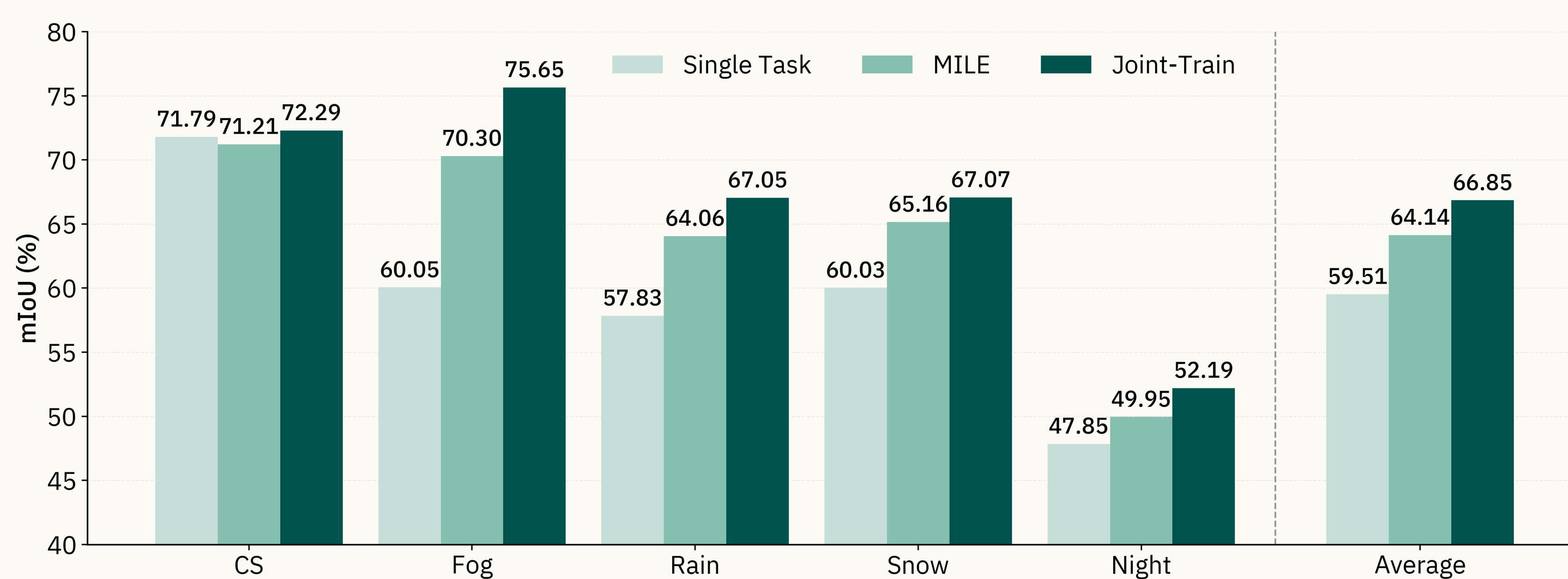
INFERENCE

Router matches scene embedding to the nearest prototype and selects the corresponding expert.

EFFICIENCY

≈5% parameter growth per task
· 1 shared backbone · 1 prototype per task

Results: Adverse Weather Domains

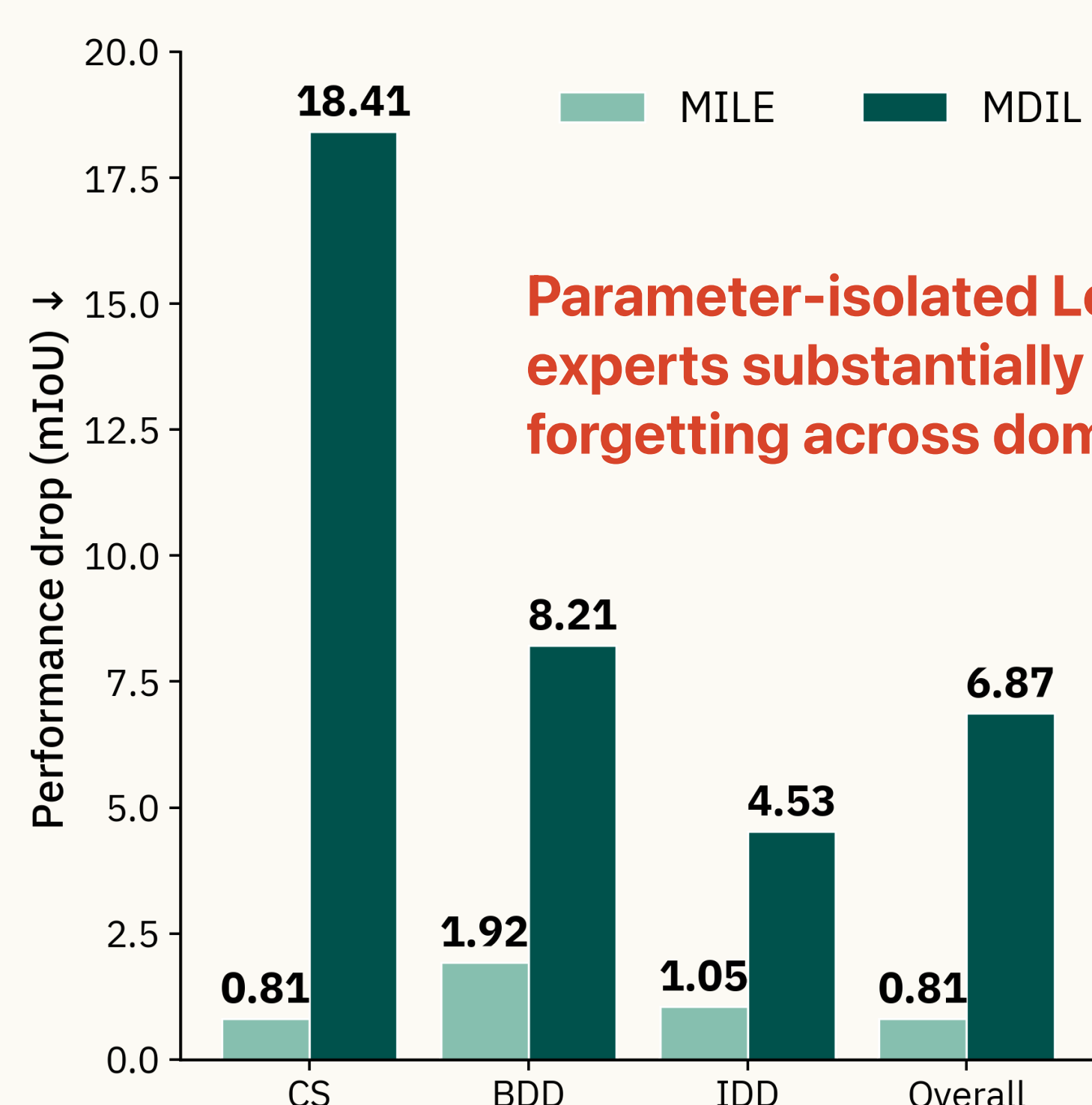


MILE improves over Single Task by +4.63 mIoU and approaches Joint Training performance.

Expert Routing

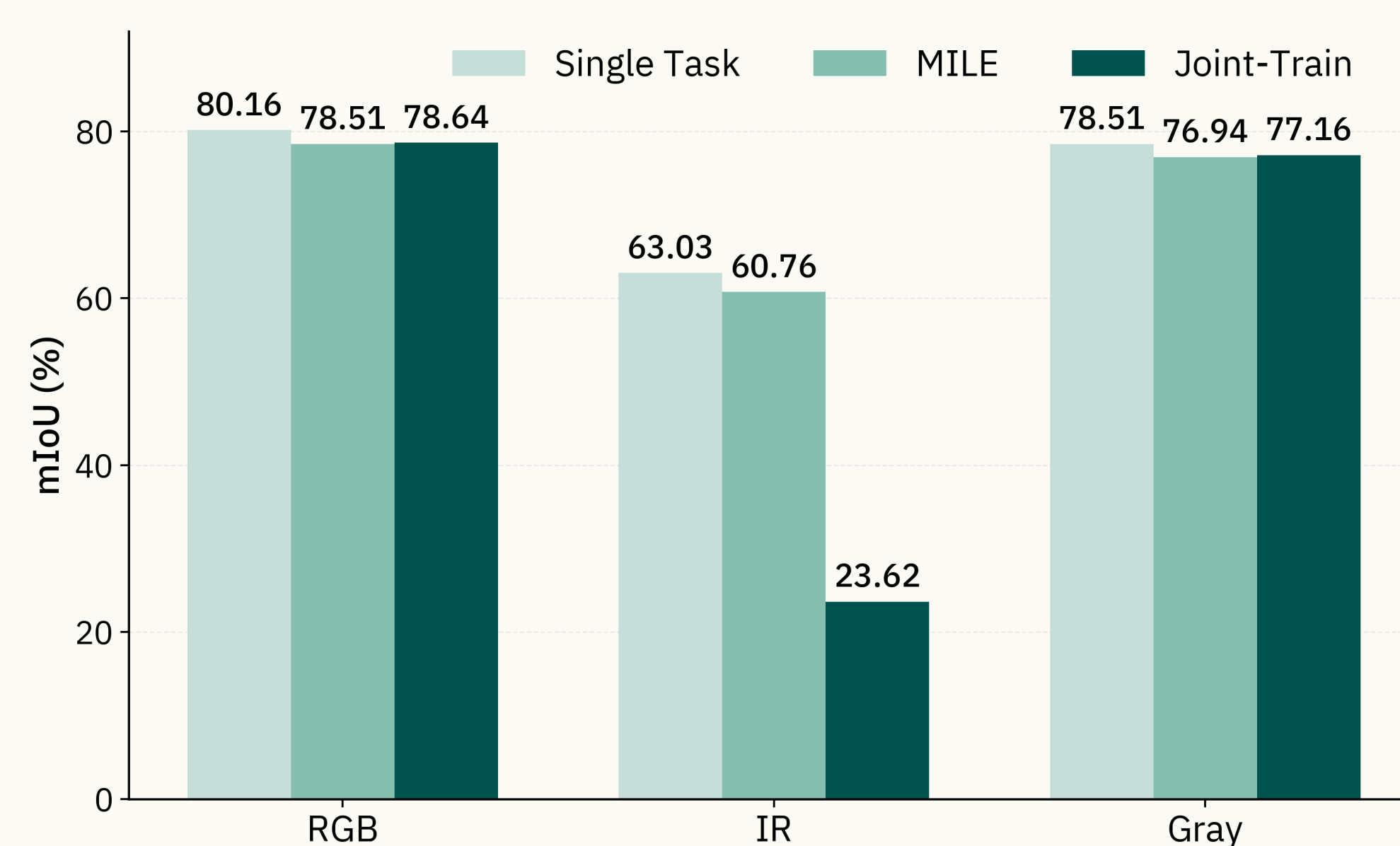
True domain	CS	100	0	0	0	0	Accuracy: 90.4%	
	Fog	0	80	3	17	0		
	Rain	0	3	85	12	0		Results (mIoU): Learned: 64.14
	Snow	0	1	12	87	0		
	Night	0	0	0	0	100		Oracle: 64.42
	CS	Fog	Rain	Snow	Night			
Predicted domain								

Resistance to Forgetting



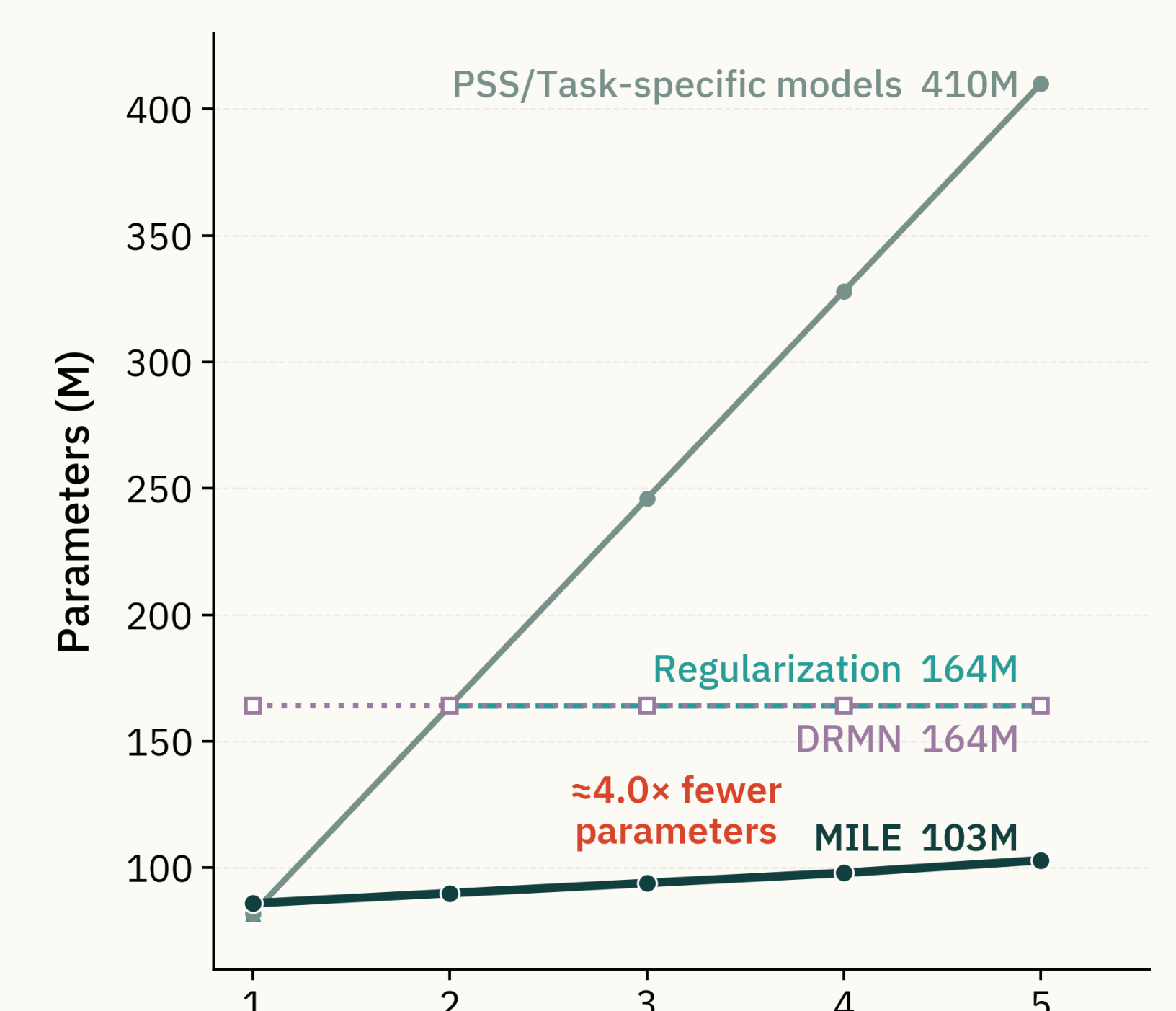
Parameter-isolated LoRA experts substantially reduce forgetting across domains.

Modality-Incremental



Large shifts favor specialization: MILE avoids negative interference observed in Joint Training.

Parameter Growth over Tasks



Contact

Shishir.Muralidhara@dfki.de

Rene.Schuster@dfki.de



References

- Garg et al. Multi-domain incremental learning for semantic segmentation, WACV 2022
- Hegde et al. Modality-incremental learning with disjoint relevance mapping networks for image-based semantic segmentation, WACV 2025
- Muralidhara et al. Domain-incremental semantic segmentation for autonomous driving under adverse driving conditions, ICPRAM 2025

Acknowledgments

This work was partially funded by the German Federal Ministry of Research, Technology, and Space under the project COPPER (16IW24009).



Federal Ministry of Research, Technology and Space